Project Phase 3: Relational Model and Normalization

Submitted by: Team 28

Assigned Phase 2 submission: Team 37 (Telomere) – https://eukonkanto.fi/en/front-page/

We converted the final EER diagram of Team 37's Phase 2 submission (their model of Team 41's wife-carrying championship requirements), including the corrections and assumptions they list in their submission. The mapping follows Elmasri & Navathe, and normalization uses the general definitions of 2NF and 3NF based on all candidate keys.

Stage	Result
A	26 relations mapped from 16 entity types (9 strong, 2 subclasses, 5 weak), 24 relationship types and 3 multi-valued attributes.
B (1NF)	No conversion needed: every attribute is already atomic after Stage A.
C (2NF)	One change: accreditation_year removed from ACCREDITS (partial dependency champ_id → accreditation_year).
D (3NF)	No conversion needed: every non-trivial dependency has a superkey on its left side, except one calculated dependency in RACE_RESULT that is set aside (Section 4).

1 Stage A: EER-to-Relational Mapping

1.1 What became which relation

EER construct	Relation(s) in Stage A	Textbook rule / option
Strong entity types PERSON, COUNTRY, TEAM, RACE_CATEGORY, CHAMPIONSHIP, VENUE, ORGANISER, SPONSOR, PRESS_OUTLET	One relation each, same name. Composite attributes become their components: full_name → first_name, last_name; VENUE.address → street, city, postal_code, country_code; contact → contact_email, contact_phone, contact_address (SPONSOR, PRESS_OUTLET).	Step 1
Specialization of PERSON into COMPETITOR and OFFICIAL (disjoint, partial, defined by role_type)	PERSON keeps role_type; COMPETITOR(person_id, bib_number, role, weight_kg, career_start_year) and OFFICIAL(person_id, certification_level, official_role), person_id = PK and FK → PERSON.	Step 8, option 8A
Weak PAIR, owners COMPETITOR via HAS_CARRIER and HAS_CARRIED, no partial key	PAIR(carrier_id, carried_id, formed_year, is_active); PK = both owner keys, each an FK → COMPETITOR.	Step 2
Weak RACE_HEAT, owners CHAMPIONSHIP (CHAMP_HEAT) and RACE_CATEGORY (CAT_HEAT), partial key heat_number	RACE_HEAT(champ_id, cat_code, heat_number, scheduled_time, status).	Step 2
Weak RACE_RESULT, owners PAIR (RESULT_OF) and RACE_HEAT (HEAT_RESULT), no partial key	RACE_RESULT, PK (carrier_id, carried_id, champ_id, cat_code, heat_number); owners mapped first.	Step 2
Weak PROTEST (owner CHAMPIONSHIP, partial key protest_seq) and PROTEST_DECISION (owner PROTEST, partial key decision_seq)	PROTEST, PK (champ_id, protest_seq); PROTEST_DECISION, PK (champ_id, protest_seq, decision_seq).	Step 2

EER construct	Relation(s) in Stage A	Textbook rule / option
1:N relationships NATIONALITY, HOME_COUNTRY, HOSTS, OWNS, FILED_BY, AGAINST_PAIR, CONCERNS_HEAT, DECIDED_BY	Foreign key on the N side: PERSON.nationality_code; TEAM.home_country_code; CHAMPIONSHIP.venue_id and organiser_id; PROTEST.(filer_carrier_id, filer_carried_id), (against_carrier_id, against_carried_id) and (champ_id, heat_cat_code, heat_number); PROTEST_DECISION.decided_by.	Step 4, foreign-key approach
Recursive 1:N TEAM_PARTNERSHIP (affiliate 0..1, parent 0..N)	TEAM.parent_team_id → TEAM.team_id.	Step 4 (recursive)
M:N relationships COMPETES_FOR, HAS_CATEGORY, PARTICIPATES_IN, SPONSORS, ACCREDITS	Relationship relations with both owners' keys as PK; relationship attributes stay in them: SPONSORS(tier, contract_start, contract_end), ACCREDITS(accreditation_year, drone_permission).	Step 5
REPRESENTS: PAIR (1,2) – COUNTRY (0,N)	REPRESENTS(carrier_id, carried_id, country_code). A pair can represent two countries, so one FK in PAIR cannot hold it.	Step 5 (relationship relation)
Ternary JUDGES: OFFICIAL, CHAMPIONSHIP, RACE_CATEGORY (all many)	JUDGES(official_id, champ_id, cat_code), PK = all three FKs.	Step 7
Multi-valued phone_numbers, host_languages, sponsorship_categories	PERSON_PHONE, CHAMP_HOST_LANGUAGE, SPONSOR_CATEGORY.	Step 6, option (b)
Union types (categories)	None in the diagram.	–

1.2 Choices and notes

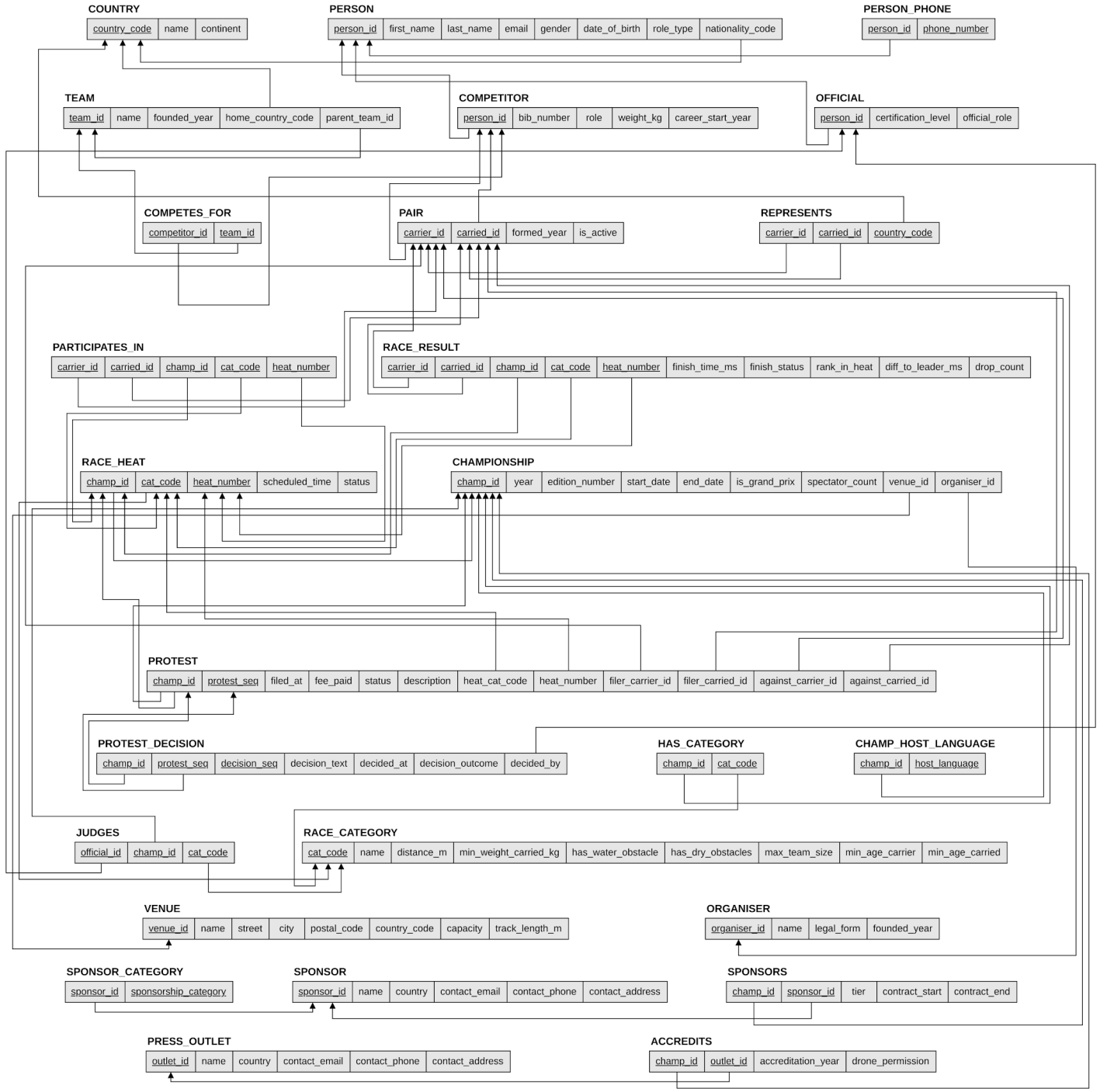
- **Subclasses (option 8A).** Both subclasses take part in their own relationships: COMPETITOR in HAS_CARRIER, HAS_CARRIED and COMPETES_FOR, OFFICIAL in JUDGES and DECIDED_BY. With a separate relation per subclass, PAIR.carrier_id, PAIR.carried_id and COMPETES_FOR.competitor_id can reference only competitors, and JUDGES.official_id and PROTEST_DECISION.decided_by only officials. Option 8C would also fit a disjoint specialization, but those foreign keys could then point at any person. The specialization is partial, so a person in neither subclass simply has no COMPETITOR or OFFICIAL tuple.
- **Multi-valued attributes, option (b) for all three.** Neither the diagram nor Team 37's notes gives a maximum number of phone numbers per person, host languages per championship or sponsorship categories per sponsor, so a fixed set of columns has no basis. Each new relation's key is the owner's key plus the value: (person_id, phone_number), (champ_id, host_language), (sponsor_id, sponsorship_category).
- **Derived attributes not stored:** PERSON.age (from date_of_birth), COUNTRY.first_participation_year (from the earliest championship of a pair representing the country), PAIR.weight_deficit_kg (from the carried competitor's weight_kg and the category's min_weight_carried_kg), CHAMPIONSHIP.years_since_inaugural (from year) and RACE_RESULT.rank_in_category (from the results of the category). rank_in_heat and diff_to_leader_ms are drawn as stored attributes and Team 37 state that they kept them stored, so they are stored.
- **Shared champ_id in PROTEST.** The protest's own champ_id (from its owner CHAMPIONSHIP) is also the first column of the CONCERNS_HEAT foreign key (champ_id, heat_cat_code, heat_number) → RACE_HEAT. Team 37 require the concerned heat to belong to the protest's championship; sharing the column enforces this without a second championship column.
- **Address and contact countries are not foreign keys.** VENUE.country_code is a component of the composite address, and SPONSOR.country and PRESS_OUTLET.country are simple

attributes. The diagram has no relationship from these entity types to COUNTRY, so we did not add one.

- **Foreign keys are renamed by role** where the same key appears twice or its meaning differs: nationality_code, home_country_code, parent_team_id, competitor_id, carrier_id / carried_id, official_id, decided_by, filer_… / against_…, heat_cat_code.
- **Participation constraints that keys cannot express.** The (1,1) participations of PERSON in NATIONALITY and of CHAMPIONSHIP in HOSTS and OWNS become NOT NULL foreign keys. The minimums RACE_HEAT (2,N) in PARTICIPATES_IN, CHAMPIONSHIP (1,N) in SPONSORS and JUDGES, RACE_CATEGORY (1,N) in JUDGES and the maximum PAIR (1,2) in REPRESENTS need CHECK constraints or triggers for later phases, as do the rules Team 37 list as notational deviations (carrier ≠ carried, jury-only decisions, results only for participating pairs).
- **Notation.** As in Elmasri & Navathe: relation name above a row of attributes, primary key underlined, and a directed arc from each foreign key to the primary key it references.
  Deviation: the textbook figures only show single-attribute foreign keys, so a composite foreign key (e.g. RACE_RESULT → PAIR, RACE_RESULT → RACE_HEAT) is drawn as one arrow per attribute, each pointing to the matching primary-key attribute.

Stage A snapshot

Relational schema after mapping Team 37's EER diagram. Primary keys are underlined; each arrow runs from a foreign-key attribute to the primary-key attribute it references.



Link: https://drive.google.com/file/d/1VukPMniTf0UCkZL0hTp5cqLYLex4I38e/view?usp=sharing

2 Stage B: Conversion to 1NF

No conversion needed; Stage B is identical to Stage A, so no separate snapshot is given.

Every attribute in the 26 Stage A relations holds one atomic value. The composite attributes full_name, address (VENUE) and contact (SPONSOR, PRESS_OUTLET) were already replaced by their simple components in Stage A, and the three multi-valued attributes are stored one value per tuple in PERSON_PHONE, CHAMP_HOST_LANGUAGE and SPONSOR_CATEGORY. No relation has a list-valued column, a nested relation or a repeating group, and no fixed column sets (as per option (a)) were used.

3 Stage C: Conversion to 2NF

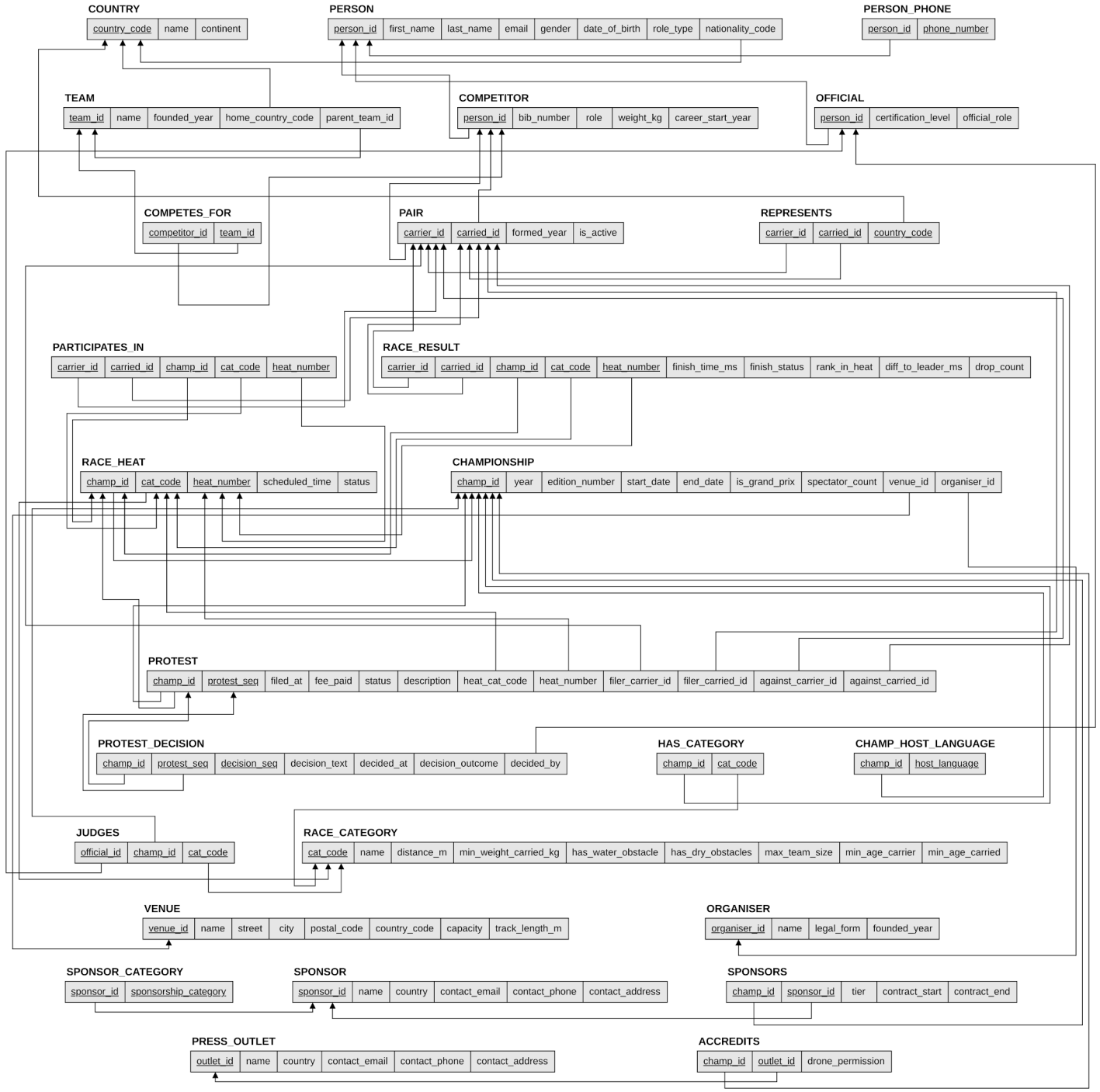
A partial dependency is only possible on a candidate key with more than one attribute, so we checked every relation with a composite candidate key (Section 5 lists all keys and dependencies). One relation violates 2NF:

- **ACCREDITS(champ_id, outlet_id, accreditation_year, drone_permission).** Its only candidate key is {champ_id, outlet_id}. In their Redundancy Analysis, Team 37 noted that “accreditation_year is redundant only if it means the year of the linked championship, rather than the year when accreditation was approved.” Their diagram gives no approval date or other timing for ACCREDITS, so we take the first meaning: accreditation_year is the year of the championship the outlet is accredited for, i.e. the value already stored as CHAMPIONSHIP.year. Under this meaning **champ_id → accreditation_year** holds. accreditation_year is non-prime and depends on champ_id, a proper subset of the key: a partial dependency, so ACCREDITS is not in 2NF. The year is repeated for every outlet accredited for the same championship and can disagree with CHAMPIONSHIP.year.
- **Decomposition.** ACCREDITS is split into ACCREDITS(champ_id, outlet_id, drone_permission) and R(champ_id, accreditation_year). R has key champ_id and records the same fact as CHAMPIONSHIP.year (it equals π champ_id, year (CHAMPIONSHIP)), so it is merged into CHAMPIONSHIP instead of being kept as a new relation; the accreditation year is obtained by joining ACCREDITS with CHAMPIONSHIP on champ_id. The split is non-additive by the NJB test: ACCREDITS ∩ R = {champ_id} and champ_id → accreditation_year. The foreign keys champ_id → CHAMPIONSHIP and outlet_id → PRESS_OUTLET are unchanged, and drone_permission still depends on the whole key, because the permission is granted per outlet per championship.

All other relations are already in 2NF. After Stage C the model still has 26 relations; only ACCREDITS changed.

Stage C snapshot

Relational schema after conversion to 2NF. The only change from Stage A is ACCREDITS, which no longer contains accreditation_year.



Link: https://drive.google.com/file/d/1g8-J0z6UjQEhZ9-izQIMjWxgIPuQ8Kfs/view?usp=sharing

4 Stage D: Conversion to 3NF

No conversion needed; Stage D is identical to Stage C, so no separate snapshot is given.

In every Stage C relation, each non-trivial functional dependency listed in Section 5 has a superkey on its left side, with one exception: the calculated dependency champ_id, cat_code, heat_number, finish_time_ms → diff_to_leader_ms in RACE_RESULT, explained below. Apart from that dependency, the only determinants are candidate keys, including the alternative keys email in PERSON and name in COUNTRY, so no non-prime attribute depends transitively on a key. Three relations needed a closer look:

- **CHAMPIONSHIP.** Only champ_id is a key: year and edition_number are not unique (several championships, such as Grand Prix events, can share them). There is no dependency among the non-key attributes. In particular start_date does not determine year, because year names the edition and a postponed edition keeps its year.
- **VENUE.** We do not treat (postal_code, country_code) → city as a functional dependency, because one postal code can cover several towns. With no dependency between non-key attributes, VENUE is in 3NF.
- **RACE_RESULT.** Within a heat, equal finishing times give equal gaps to the leader, so champ_id, cat_code, heat_number, finish_time_ms → diff_to_leader_ms holds. diff_to_leader_ms is a calculated value: finish_time_ms minus the fastest finish_time_ms of the same heat, read from other tuples. Elmasri & Navathe note that dependencies defined by such arithmetic functions or procedures are technically present but are not given particular attention during normalization. Splitting RACE_RESULT on it would create a relation keyed by heat and finishing time that stores the same subtraction and must still be updated whenever the heat's fastest time is corrected, so it would not remove the update anomaly Team 37 described. We therefore list this dependency and leave RACE_RESULT unchanged. Strictly, this dependency fails the 3NF test: its left side is not a superkey and diff_to_leader_ms is non-prime. It is the only such dependency in the model, and we leave it in place for the reason above. rank_in_heat is not determined by finish_time_ms, because finish_status (DNF, DQ) and the tie rules also decide it.

5 Functional Dependencies and Candidate Keys

The dependencies come from the meaning of the attributes in Team 37's diagram and notes. Trivial dependencies are omitted, and a relation whose key is all of its attributes (all-key) has no non-trivial dependencies. A prime attribute belongs to some candidate key. The dependency that caused a change is **champ_id → accreditation_year** in ACCREDITS (Stage C).

Relation	Candidate keys	Functional dependencies checked	Notes and result
PERSON	{person_id}, {email}	person_id → first_name, last_name, email, gender, date_of_birth, role_type, nationality_code; email → person_id	Prime: person_id, email. Both determinants are keys: 3NF
PERSON_PHONE	{person_id, phone_number}	none (all-key)	3NF
COMPETITOR	{person_id}	person_id → bib_number, role, weight_kg, career_start_year	bib_number is unique only within a championship, so not a key here. 3NF
OFFICIAL	{person_id}	person_id → certification_level, official_role	3NF
COUNTRY	{country_code}, {name}	country_code → name, continent; name → country_code	Country names are unique. 3NF
TEAM	{team_id}	team_id → name, founded_year, home_country_code, parent_team_id	3NF
COMPETES_FOR	{competitor_id, team_id}	none (all-key)	3NF
PAIR	{carrier_id, carried_id}	carrier_id, carried_id → formed_year, is_active	A competitor is in many pairs, so neither id alone determines them. 3NF

Relation	Candidate keys	Functional dependencies checked	Notes and result
REPRESENTS	{carrier_id, carried_id, country_code}	none (all-key; a pair may represent two countries)	3NF
VENUE	{venue_id}	venue_id → name, street, city, postal_code, country_code, capacity, track_length_m	(postal_code, country_code) → city not assumed. 3NF
ORGANISER	{organiser_id}	organiser_id → name, legal_form, founded_year	3NF
CHAMPIONSHIP	{champ_id}	champ_id → year, edition_number, start_date, end_date, is_grand_prix, spectator_count, venue_id, organiser_id	year, edition_number not unique; start_date ↛ year. 3NF
CHAMP_HOST_LANGUAGE	{champ_id, host_language}	none (all-key)	3NF
RACE_CATEGORY	{cat_code}	cat_code → name, distance_m, min_weight_carried_kg, has_water_obstacle, has_dry_obstacles, max_team_size, min_age_carrier, min_age_carried	3NF
HAS_CATEGORY	{champ_id, cat_code}	none (all-key)	3NF
JUDGES	{official_id, champ_id, cat_code}	none (all-key; an official judges many categories, a category has many officials)	3NF
RACE_HEAT	{champ_id, cat_code, heat_number}	champ_id, cat_code, heat_number → scheduled_time, status	heat_number restarts per championship and category, so no subset of the key determines them. 3NF
PARTICIPATES_IN	{carrier_id, carried_id, champ_id, cat_code, heat_number}	none (all-key)	3NF
RACE_RESULT	{carrier_id, carried_id, champ_id, cat_code, heat_number}	key → finish_time_ms, finish_status, rank_in_heat, diff_to_leader_ms, drop_count; champ_id, cat_code, heat_number, finish_time_ms → diff_to_leader_ms (calculated value)	No partial dependency (a pair has many results, a heat has many results). Calculated FD listed, not decomposed (Section 4).
PROTEST	{champ_id, protest_seq}	champ_id, protest_seq → filed_at, fee_paid, status, description, heat_cat_code, heat_number, filer_carrier_id, filer_carried_id, against_carrier_id, against_carried_id	champ_id alone determines none (many protests per championship); fee_paid is recorded per protest. 3NF
PROTEST_DECISION	{champ_id, protest_seq, decision_seq}	champ_id, protest_seq, decision_seq → decision_text, decided_at, decision_outcome, decided_by	A protest can have several decisions (0,N), so (champ_id, protest_seq) determines none. 3NF
SPONSOR	{sponsor_id}	sponsor_id → name, country, contact_email, contact_phone, contact_address	3NF
SPONSOR_CATEGORY	{sponsor_id, sponsorship_category}	none (all-key)	3NF
SPONSORS	{champ_id, sponsor_id}	champ_id, sponsor_id → tier, contract_start, contract_end	tier and dates belong to the contract, not to the sponsor or championship alone. 3NF
PRESS_OUTLET	{outlet_id}	outlet_id → name, country, contact_email, contact_phone, contact_address	3NF

Relation	Candidate keys	Functional dependencies checked	Notes and result
ACCREDITS (Stages A, B)	{champ_id, outlet_id}	champ_id, outlet_id → accreditation_year, drone_permission; **champ_id → accreditation_year**	accreditation_year non-prime, partial dependency: not 2NF, decomposed in Stage C
ACCREDITS (Stages C, D)	{champ_id, outlet_id}	champ_id, outlet_id → drone_permission	3NF

6 Changes to the Assigned EER Diagram

None. We mapped Team 37's final diagram as drawn, including the corrections and assumptions they list. The limitations they record themselves, such as registration and team rosters without championship context, the global disjointness of COMPETITOR and OFFICIAL, and a single weight_kg per competitor, are gaps in Team 41's requirements rather than errors in EER notation.

7 Use of Generative AI

We used generative AI while preparing this phase. It read Team 37's Phase 2 submission, proposed the ER/EER-to-relational mapping and the first draft of the explanations. We then used draw.io to draw the relational schema based on the outline provided by gen AI. The AI had some open questions, we decided: that accreditation_year is the championship's year, that postal codes are not taken to determine cities, that CHAMPIONSHIP.year and edition_number are not unique, and that the calculated diff_to_leader_ms is listed but not decomposed. We checked every relation, key, foreign key and dependency against Team 37's diagram and the textbook. We drafted the final version of this report.